RAJ DANGI

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SUMMARY

Computer science student at the University of Toledo. I work on data and reporting, build web systems, and am currently focused on understanding AI. First-author paper on high-availability web systems accepted for oral presentation at ISDFS 2026.

EDUCATION

University of Toledo · *B.S. Computer Science* — Aug 2023 – May 2027

Relevant coursework: Computer Architecture, Operating Systems, Data Communications, Network Security, Data Structures.

EXPERIENCE

Sports Analytics & Data Science Intern · *University of Toledo Athletics* — 2026

- Build internal reporting tools for the Athletics marketing team.
- Maintain the data process behind the reports so they stay accurate and up to date.
- Work with staff to turn their recurring questions into reports.

Software Developer · *Batting Cleanup (Civic-Tech Initiative)* — Aug 2025 – Apr 2026

- Built a QR-code reporting system that replaced manual tracking and cut each submission to under 5 seconds.
- Added usage analytics with PostHog to find where the app was slow or confusing.

Amazon DeepRacer Competitor · *University of Toledo* — Jan 2025

- Placed in the top 5 of 60+ participants by writing reward functions for reinforcement-learning models on AWS.

PROJECTS

Iris: High-Availability Web Stack — HAProxy, Linux, Tailscale

github.com/rajdangi31/iris-lamp

- Designed a two-node web stack that switches to a backup server when the main one fails. First-author paper accepted for oral presentation at ISDFS 2026.
- Measured about 3 s to fail over and 5 s to fail back, with almost no added latency. Servers connect over Tailscale and sync with rsync and mysqldump.

NYXIS: AI Productivity App — PostgreSQL, TypeScript, Supabase

github.com/rajdangi31/NYXIS

- Built a productivity app that turns personal goals into AI-generated quests and tracks progress over time, backed by PostgreSQL.

GPT-2 Transformer from Scratch — PyTorch, Transformers

github.com/rajdangi31/GPT-2

- Wrote a GPT-2 style language model from scratch, including attention, the tokenizer, and the training loop.

TakeAPeek: Social App — React, Supabase

github.com/rajdangi31/takeApeek

- Built a social app for sharing posts, with sign-in, real-time updates, and push notifications.

SKILLS

Languages	Python, TypeScript, JavaScript, C, C++, Java, SQL
Systems	Linux, Docker, networking, high-availability and failover design, Git
Data	PostgreSQL, Supabase, data pipelines, validation, reporting and dashboards
AI / ML	PyTorch, transformers, retrieval-augmented generation, LLM evaluation, reinforcement learning